

# Mrinaal Dogra

MASTERS IN COMPUTER SCIENCE · UC SAN DIEGO

☎ +1 (858) 518-3496 | ✉ mdogra@ucsd.edu | 🏠 mrinaald.github.io | 📷 mrinaald | 📺 mrinaald

## Work Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Kognitos</b><br>MACHINE LEARNING ENGINEER INTERN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | San Jose, CA<br>Jun. 2025 - Sep. 2025     |
| <ul style="list-style-type: none"><li>Researched and prototyped an <b>agentic AI pipeline</b> for <b>SOP generation</b> from desktop recordings, integrating key-frame extraction (CV-based), OCR, and multi-agent reasoning; benchmarked GPT-4o, GPT-4.1, GPT-5, and Gemini-2.5.</li><li>Built and deployed a lightweight <b>multimodal web app</b> (video, images, documents) for automated <b>SOP and flow-diagram generation</b>, <b>containerized with Docker</b> and hosted on <b>AWS EC2</b>; enabled internal testing across diverse inputs.</li><li>Developed and productionized a <b>PII masking solution</b> using <b>GLiNER</b> model (supports 40+ entity types), evolving from a <b>REST API</b> on <b>AWS EC2</b> (PoC) to a scalable <b>AWS Lambda</b> service with Docker packaging and role-based access control.</li></ul> |                                           |
| <b>Samsung R&amp;D Institute India - Bangalore</b><br>LEAD ENGINEER, MACHINE LEARNING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Bangalore, India<br>Mar. 2023 - Aug. 2024 |
| <ul style="list-style-type: none"><li>Led the development of an <b>automated Python-based Android profiling tool</b> to benchmark rendering uniformly across devices, identifying bottlenecks (scroll janks) and enhancing rendering performance through improvements in the Android framework source.</li><li>Led the development of <b>four edge-based personalization LLM solutions</b>, deriving actionable insights to enhance user experience.</li><li>Fine-tuned the <b>FLAN-T5 LLM model</b> for our specific use-case, optimizing performance &amp; achieving a <b>~ 92% evaluation score</b>.</li><li>Collaborated with cross-functional teams to improve the quality of <b>~ 15k samples</b> for one of the said personalization solutions.</li></ul>                                                                              |                                           |
| <b>SENIOR SOFTWARE ENGINEER, MACHINE LEARNING</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Mar. 2021 - Feb. 2023                     |
| <ul style="list-style-type: none"><li>Designed an <b>edge ML solution</b> to analyze phone usage data and <b>detect boredom</b> with <b>~ 80% accuracy</b>, enhancing user experience. Built an Android app for <b>real-time inference with under 50ms latency</b>, showcasing the model's effectiveness.</li><li>Pioneered a <b>Federated Learning (FL)</b>-based solution to <b>predict gender and demographic age</b>, <b>enhancing privacy</b> for <b>~ 10k users</b>. Explored innovative distributed learning techniques and tested diverse FL algorithms across 20+ input and model configurations, laying the groundwork for future privacy-preserving AI advancements.</li><li>Developed a <b>differential privacy-based ML solution</b> for assigning semantic tags to locations frequently visited by a user.</li></ul>            |                                           |
| <b>SOFTWARE ENGINEER, MACHINE LEARNING</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Jun. 2019 - Feb. 2021                     |
| <ul style="list-style-type: none"><li>Developed the Robot Camera Visualization Android app for <b>real-time visualization</b> of depth maps and 3D point-clouds from a ToF camera at <b>30 FPS</b>, with gesture-based UI features for enhanced user interaction, tailored to stakeholder requirements.</li><li>Engineered a <b>privacy-preserving edge ML solution</b> for <b>Next App Recommendation</b>, published in <b>IEEE ICSC 2022</b>, by designing a memory-efficient model (<b>99% size reduction</b>) to minimize FL bandwidth costs. Trained and deployed the model in Java using <b>DL4J</b>, integrating it on Android edge devices with a User Trial (UT) app for <b>training and inference across 500+ devices</b></li></ul>                                                                                                 |                                           |
| <b>Samsung R&amp;D Institute India - Bangalore</b><br>UNDERGRADUATE SOFTWARE DEVELOPER INTERNSHIP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Bangalore, India<br>May 2018 - Jul. 2018  |
| <ul style="list-style-type: none"><li>Designed and developed a <b>neural network model</b> to <b>predict a user's location</b> based on recent travel data and time of day.</li><li>Created a <b>Python-based simulation environment</b> to model cell tower connectivity while a user is in transit.</li><li>Built a <b>machine learning classification model</b> to predict the most likely cell tower a user is connected to at any given time, achieving <b>Top-1 and Top-3 accuracies of 85-90% and 90-95%</b>, respectively, on the in-house evaluation dataset.</li></ul>                                                                                                                                                                                                                                                              |                                           |
| <b>Hike Private Limited</b><br>UNDERGRADUATE SOFTWARE DEVELOPER INTERNSHIP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | New Delhi, India<br>May 2017 - Jul. 2017  |
| <ul style="list-style-type: none"><li>Implemented <b>CNN</b> models in Python using <b>TensorFlow</b> for <b>image classification</b>, leveraging Google ML Engine APIs to accelerate training on Google Cloud. Deployed the model with <b>TensorFlow Serving</b> to expose REST APIs for generating model predictions.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                           |

## Skills

|                             |                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------|
| <b>Programming</b>          | Python, JAVA, C/C++, Shell Script(Bash), Go, LaTeX                                               |
| <b>Libraries</b>            | Tensorflow, PyTorch, Scikit-learn, DeepLearning4Java(DL4J), Flask, MPICH C++, OpenCV, CUDA C/C++ |
| <b>Software &amp; Tools</b> | Git, Perforce Helix Core, GDB, ROS                                                               |
| <b>OS and Platforms</b>     | Linux, Windows, Raspberry-Pi, Arduino                                                            |
| <b>Languages</b>            | English(Fluent), Hindi(Native Speaker)                                                           |

## Education

| Year         | Degree                                  | Institute    | CPI/%    |
|--------------|-----------------------------------------|--------------|----------|
| 2024-Present | Masters in Computer Science             | UC San Diego | 3.96/4.0 |
| 2015-2019    | B.Tech Computer Science and Engineering | IIT Kanpur   | 9.0/10.0 |

# Patents and Publications

---

## PUBLICATIONS

**Memory Efficient Federated Recommendation Model** [\[link\]](#) 2022  
2022 IEEE 16TH INTERNATIONAL CONFERENCE ON SEMANTIC COMPUTING (ICSC)

## PATENTS

**System and Method for Distributed Learning of Universal Vector Representations on Edge Devices** [\[link\]](#) 2023  
US 17/946349

**Methods and Electronic Devices for Behavior Detection using Federated Learning** [\[link\]](#) 2023  
US 18/191403

**One more patent** has been filed and is in the publication process 2023

# Awards and Achievements

---

|      |                                                                                                                                              |            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 2024 | <b>Samsung Excellence Award</b> , Recognized as <b>Star of the Quarter</b> for excellent contributions in projects                           | SRI-B      |
| 2023 | <b>Key Talent Recognition Program</b> , Recognized for exemplary teamwork and significant contributions to projects and organizational goals | SRI-B      |
| 2022 | <b>SPOT Award</b> , Recognized for excellent contributions in the project                                                                    | SRI-B      |
| 2021 | <b>SPOT Award</b> , Recognized for swift implementation and completion of project                                                            | SRI-B      |
| 2020 | <b>Clean Code Culture Award</b> , Recognized for rigorous adoption of clean code and SOLID principles in project                             | SRI-B      |
| 2020 | <b>Samsung Citizenship Award</b> , Recognized for excellent contributions and commitment to project                                          | SRI-B      |
| 2018 | <b>Academic Excellence Award</b> , 2017-18 Academic Year, Dept. of Computer Science and Engineering                                          | IIT Kanpur |

# Relevant Coursework

---

## GRADUATE

|                                |                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Artificial Intelligence</b> | CSE250A: Principles of AI: Probabilistic Reasoning and Learning<br>CSE 291A: Large Model Reasoning<br>CSE 258: Web Mining and Recommender Systems<br>CSE 256: Statistical Natural Language Processing ( <i>ongoing</i> )<br>CSE 257: Search and Optimization ( <i>ongoing</i> )<br>CSE 291A: AI Agents ( <i>ongoing</i> ) |
| <b>ML System</b>               | CSE 234: Data Systems for Machine Learning<br>CSE 291A: Systems for LLMs and AI Agents                                                                                                                                                                                                                                    |
| <b>Others</b>                  | CSE 202: Algorithm Design and Analysis<br>CSE 224: Graduate Networked Systems<br>CSE 276C: Mathematics for Robotics                                                                                                                                                                                                       |

## ONLINE COURSES

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Machine Learning</b> | Generative AI with Large Language Models - DeepLearning.AI, Coursera<br>Reinforcement Learning Specialization - University of Alberta, Coursera (4 Courses)<br>Generative Adversarial Networks (GANs) Specialization - DeepLearning.AI, Coursera (3 Courses)<br>Machine Learning Engineering for Production (MLOps) Specialization - DeepLearning.AI, Coursera (4 Courses)<br>Convolutional Neural Networks - DeepLearning.AI, Coursera<br>Sequence Models - DeepLearning.AI, Coursera<br>Hyperparameter Tuning, Regularization and Optimization - DeepLearning.AI, Coursera |
| <b>Robotics</b>         | Robotics Specialization - University of Pennsylvania, Coursera (6 Courses)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## UNDERGRADUATE

|                         |                                                                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Machine Learning</b> | Introduction to Machine Learning, Natural Language Processing, Data Mining, Computational Cognitive Science                                                                                                        |
| <b>Computer Science</b> | Operating Systems, Computer Networks, Parallel Computing, Data Structure & Algorithm, Advanced Algorithms, Compiler Design, Computer Systems Security, Introduction to Software Engineering, Blockchain Technology |
| <b>Others</b>           | Introduction to Electronics, Introduction to Electrical Engineering, Neurobiology                                                                                                                                  |

## Positions of Responsibility

|         |                                                                                                                 |            |
|---------|-----------------------------------------------------------------------------------------------------------------|------------|
| 2025    | <b>Teaching Assistant</b> , For DSC 140: Probabilistic Modeling and ML, by Prof. Tevfik Berk Ustun, Fall 2025   | UCSD       |
| 2025    | <b>Teaching Assistant</b> , For DSC 140: Probabilistic Modeling and ML, by Prof. Tevfik Berk Ustun, Spring 2025 | UCSD       |
| 2025    | <b>Teaching Assistant</b> , For DSC 240: Introduction to Machine Learning, by Prof. Yu-Xiang Wang, Winter 2025  | UCSD       |
| 2024    | <b>Project Lead</b> , Supervised and contributed to four different on-device personalization solutions          | SRI-B      |
| 2023    | <b>Project Lead</b> , Pioneered and supervised the development of the in-house Android profiling tool           | SRI-B      |
| 2022    | <b>Project Mentor</b> , Worked with and mentored two Interns in two different projects                          | SRI-B      |
| 2019    | <b>Project Mentor</b> , Project under Association of Computer Activities (ACA)                                  | IIT Kanpur |
| 2018    | <b>Teaching Assistant</b> , For ESO 207A: Data Structures and Algorithms, by Prof. Sumit Ganguly                | IIT Kanpur |
| 2017-18 | <b>Event Manager</b> , Robogames Techkriti'18                                                                   | IIT Kanpur |
| 2016-17 | <b>Student Guide</b> , Counselling Service                                                                      | IIT Kanpur |
| 2016-17 | <b>Secretary</b> , Robotics Club                                                                                | IIT Kanpur |

## Projects

### Multi-task Learning with ToolkenGPT Framework

UC San Diego

COURSE PROJECT: LARGE MODEL REASONING , PROF. LIANHUI QIN

Sept. 2024 - Dec. 2024

- Re-implemented ToolkenGPT framework, optimizing for computational efficiency and adapting it to smaller models like Llama-3.2 1B.
- Led experimental research on multi-task capabilities, evaluating joint-task performance across different computational tasks.

### HealthCare DApp

IIT Kanpur

COURSE PROJECT: BLOCKCHAIN TECHNOLOGY AND APPLICATIONS, PROF. SANDEEP SHUKLA

Jan. 2019 - Apr. 2019

- Decentralized Application (DApp) implemented using Ethereum Blockchain platform to keep patients' and doctors' data at a health center.
- Application allows a patient to maintain their medical data and reports securely using blockchain technology.
- Patients were given full control over their data, and only they had the power to grant access of their data to any doctor.
- Application also supported appointment bookings, where a patient can book an appointment with a doctor one-week in advance.

### Multi-user P2P Video Conferencing Web Application

IIT Kanpur

COURSE PROJECT: COMPUTER NETWORKS, PROF. DHEERAJ SANGHI

Aug. 2018 - Nov. 2018

- Implemented a video conferencing web application capable of handling multi-stream video feeds of many users.
- Application supported peer-to-peer communication and multiple conference rooms, each room capable to run an independent conference.
- Used WebRTC communication technology to enable real-time media communication between peers connected in a conference room.

### One-Shot Learning

IIT Kanpur

COURSE PROJECT: DATA MINING, PROF. ARNAB BHATTACHARYA

Aug. 2018 - Nov. 2018

- One-Shot learning tries to solve the object categorization problem while using one, or only a few, samples of each of the output category/class.
- Studied and implemented the state-of-the-art methods of one shot learning, specifically Siamese networks and Matching Networks.
- Used the Omniglot and MNIST datasets for analyzing the effectiveness of implemented methods.

### Neural Network Based Modelling and Control of Quadrotor

IIT Kanpur

UNDERGRADUATE PROJECT, PROF. INDRANIL SAHA

Jan. 2018 - Apr. 2018

- Implemented Neural networks to learn a quadrotor model and its dynamics, and used them to synthesize its controller.
- Simulation environment was set up comprising of Mavros, PX4 and Gazebo which was used to fly a virtual quadrotor and to collect data.
- Collected data for multiple trajectories involving straight lines, sinusoidal, and random trajectories, for training the models.
- Models were tested against circular trajectory which was not part of the training data, and satisfactory results were obtained.

### Detecting Semantically Similar Questions on Quora Dataset

IIT Kanpur

COURSE PROJECT: NATURAL LANGUAGE PROCESSING, PROF. HARISH KARNICK

Jan. 2018 - Apr. 2018

- Performed literature review on the existing work for detecting semantically equivalent questions from any publicly available corpora.
- Implemented a state-of-the-art work and conducted hyper-parameter tuning for training the model on Quora dataset.
- Implemented another model using Siamese neural network architecture and achieved near state-of-the-art accuracy.
- Proposed and tested few variations of the Siamese network approach while trying to improve the test accuracy.
- Analyzed the effect of including few linguistic constraints in order to improve performance and analyzed the results.

### Ada to MIPS Compiler implemented in C++

IIT Kanpur

COURSE PROJECT: COMPILER DESIGN, PROF. SUBHAJIT ROY

Jan. 2018 - Apr. 2018

- Implemented an Ada to MIPS compiler using C++ as the source language of the compiler.
- Implemented language features include Basic Arithmetic operations, Range Operator, Constant Variables, Fixed size Arrays with upto two dimension support, If-Else and If-Else-Else conditionals, Switch cases, Simple for, while, and do-while loops, Procedures(Functions) and Recursions, Packages(Classes) supporting any number of data members as well as objects of other packages, and Package level Methods.
- Basic Integer and Character data types were supported for all implemented features.

## **Real-time Sentiment Analysis of Video Feed**

*IIT Kanpur*

COURSE PROJECT: INTRODUCTION TO MACHINE LEARNING, PROF. PURUSHOTTAM KAR

*Aug. 2017 - Nov. 2017*

- Analyzed performance of existing standard CNN networks like LeNet and MobileNet to classify user sentiment from real-time video feed.
- Proposed and implemented a smaller version of AlexNet in order to reduce model complexity.

## **Humanoid Robotics Project**

*IIT Kanpur*

CORE MEMBER, ROBOTICS CLUB

*Oct. 2015 - Apr. 2017*

- Implemented various algorithms such as Line Following, Object Detection and Object Tracking using OpenCV C++.
- Implemented the Speech Recognition, Chat-bot and the core system modules for the project HURO in the SnT Summer Camp 2016.
- Actively worked with team on Computer Vision problem statements required for participation in the competition HuroCup Fira, a robotic game and robotics benchmark problem for humanoid robots.
- Implemented Histogram Backprojection algorithm using OpenCV for improving object detection module of the robot.

## **N-Body Simulation in CUDA**

*IIT Kanpur*

PROJECT UNDER ACA (ASSOCIATION OF COMPUTER ACTIVITIES)

*Jan. 2016 - Apr. 2016*

- Implemented a simulation of dynamical system consisting of a large number of particles, moving under the influence of gravity
- Used the C++ CUDA APIs for parallel implementation of the simulation on a Nvidia GPU
- Implemented the visual realization of the simulation using the OpenCV library in C++